

KALPAK KANUNGO

Generative AI & Agentic AI Engineer

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CORE SKILLS

Generative AI & LLMs	Generative AI, Large Language Models (LLMs), Claude (Anthropic), Prompt Engineering, Context Grounding, AWS Bedrock
RAG & Retrieval	Retrieval-Augmented Generation (RAG), Document Ingestion & Parsing, Chunking, Embeddings, Vector Databases, Semantic Search, Retrieval Pipelines
Agentic AI	AI Agents, Multi-Agent Systems, Agent Orchestration, Tool / Function Calling, LangChain, LangGraph, Workflow Automation, Human-in-the-Loop Design
Programming	Python
Enterprise QMS Platforms	TrackWise (Complaint Handling), ValGenesis VLMS
Practices	Agile / Scrum, SDLC, Change Control, QA / UAT Testing

CERTIFICATIONS

- Claude Certified Architect – Foundations (Anthropic)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- AWS Certified Cloud Practitioner
- Cognizant Vibe Code Hackathon

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions, Kolkata, India Sept 2023 – Present

I. Generative AI & Agentic AI Engineer May 2025 – Present

Self-initiated GenAI projects built for the enterprise QMS program, alongside primary TrackWise role.

Project 1: DS Traceability Copilot – Agentic Slack Assistant

Tech Stack: Python, LangGraph, Google Gemini, ChromaDB, Sentence-Transformers, Slack Bolt, FastAPI, Google Drive API, SQLite

- Built an **end-to-end agentic AI pipeline** that links Slack user stories to **Design Specification (DS) IDs** in a **GxP-regulated** TrackWise complaint-handling document — collapsing a manual, multi-step traceability task into a **single Slack interaction**.
- Designed a **multi-agent LangGraph pipeline** (matcher via **ChromaDB + Sentence-Transformers**, a **Gemini** drafting agent that writes house-style DS entries when no match exists, and a **self-correction reviewer agent**) — improving entry **consistency** and reducing reviewer effort to a **single approve / edit decision**.
- Added a durable **human-in-the-loop approval gate** (LangGraph interrupt/resume + **SQLite checkpointing**) with **python-docx** write-back to Google Drive, using revision history as a **21 CFR Part 11-aligned audit trail** — keeping every AI-proposed change **reviewed, attributed, and audit-ready**.
- **Impact:** Automated traceability across **323 DS entries** spanning **26 configuration families**, removing recurring manual lookup and cross-referencing effort each release.

Project 2: RAG-Based Support Chatbot for Business Users

Tech Stack: Python, LangChain, Ollama (LLM), Streamlit (UI), RAG, Embeddings, Vector Database

- Turned a recurring support burden — workflow / process queries (records not moving state, fields not editable) — into **self-serve answers**, letting business users resolve common issues **in chat instead of raising tickets**.
- Built a **RAG chatbot** grounded in the project's full documentation set (**SOPs, work instructions, DS, DF sheets**), returning answers tied to **approved sources** rather than generic model knowledge — reducing **misinformation risk**.
- Engineered the **document ingestion, chunking, embedding, and retrieval pipeline** that keeps every response **traceable to source content**.
- **Impact:** Reduced process-related tickets by **47%**, cutting repetitive support load and freeing the team for higher-value work.

II. TrackWise Developer Dec 2023 – Present

Project: Enterprise QMS Modernization Program | Client: Philips | Domain: Healthcare

- **Configuration & development:** Designed and customized **12+ TrackWise workflows** and business rules for the Complaint Handling module, aligned to client SOPs and compliant with **21 CFR Part 11**.
- **Requirements:** Led **20+ requirement workshops** with global business users and quality teams, translating needs into design specifications with **zero requirement-change rework** after sign-off.
- **Validation (ValGenesis):** Managed the validation lifecycle for **3+ years**; authored and reviewed **10+ IQ / PIQ / MP documents** and Design Specifications, and led **IQ / PIQ execution** for multiple major releases with a **100% first-pass compliance record** in audits.
- **Testing & deployment:** Coordinated QA / UAT for **6+ feature deployments (40+ defects resolved before go-live)** and managed controlled releases across Dev, QA, and Production with **zero unplanned downtime**.
- **Support & upgrades:** Provided post-production support and contributed to major TrackWise upgrades.

EDUCATION

B.Tech, Computer Science and Technology, University of Engineering and Management, Kolkata

10th & 12th – Julien Day School — ISC (83%), 2019 • ICSE (90%), 2017

2019 – 2023